Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
BCA Degree Examination - APRIL 2021

AMTCA302 – NUMERICAL METHODS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Show that $\nabla = 1 - E^{-1}$.
2. What are the advantages of central difference interpolation formula?
3. Define divided difference.
4. Compare Gauss elimination method and Gauss Jordan method.
5. State modified Euler algorithm to solve $y' = f(x, y), y(x_0) = y_0$ at $x = x_0 + h$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find the difference of the polynomial $x^3 - 3x^2 + 5x + 7$ taking $h = 2$.
7. Apply Gauss forward formula to obtain $f(3.5)$ from the table below.

x	2	3	4	5
f(x)	2.626	3.454	4.784	6.986

8. Using Lagrange's formula of interpolation, find $y(10)$.

5	6	9	11
12	13	14	16

9. Solve the system of equation by Gauss elimination method
$2x + 3y - z = 5, 4x + 4y - 3z = 3, 2x - 3y + 2z = 2$

10. Using Euler's method, Solve the equation $\frac{dy}{dx} = x + y$, $y(0) = 1$, for $(0.2), (0.4), (0.6), (0.8)$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Find y at $x = 28$ using Newton's backward formula from the table.

x	20	23	26	29
y	0.3420	0.3907	0.4384	0.4848

(or)

b) Find $y(-1)$, if $y(0) = 2, y(1) = 9, y(2) = 28, y(3) = 65, y(4) = 126, y(5) = 217$.

12. a) Using stirling's formula, find $y(1.22)$ from the following table.

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8
y	0.84147	0.89121	0.93204	0.96356	0.98545	0.99749	0.99957	0.99385	0.97385

(or)

b) Given the following table, find $y(35)$ by using Bessel's formula.

x	20	30	40	50
y	512	439	346	243

13. a) Using polynomial of the third degree, complete the record given below of the export of a certain commodity during five years.

Year	1917	1918	1919	1920	1921
Exports (in tons)	443	384	-	397	467

(or)

b) Using Newton's formula for reversion of series method, find x for $y = 3000$ from the following data:

X	10	15	20
f(x)	1754	2648	3564

14. a) Solve by crout's method the following $2x + 3y + z = -1,\ 5x + y + z = 9$
$3x + 2y + 4z = 11$

(or)

b) Solve the following system of equations by using Gauss –seidel method
$8x - 3y + 2z = 20, 4x + 11y - z = 33, 6x + 3y + 12z = 35.$

15. a) Using Adam's method find $y(0.6)$ given $\frac{dy}{dx} = x + y, y(0) = 1$ with $h = 0.2$.

(or)

b) Compute $y(0.8)$ given $\frac{dy}{dx} = y - x^2, y(0.6) = 1.7379$ using R.K method fourth order.

****